GURWINDER SINGH

IT Support Engineer & Developer

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PORTFOLIO

PROFESSIONAL SUMMARY

I'm an IT Support Engineer based in Janakpuri East, New Delhi, with 2+ years of hands-on experience in desktop support, network administration, VoIP deployment, and Windows and Linux system administration. I currently keep end users and infrastructure running at Cromptech Integrated Solutions — provisioned, documented and online.

TECHNICAL SKILLS

Desktop & End-User Support • Windows 10 / 11 & Windows Server • Active Directory & User Lifecycle • TCP/IP, LAN / WAN, DHCP, DNS & VLANs • VoIP & IP Telephony (Asterisk, PJSIP, SIP / RTP) • Linux System Administration (Rocky Linux, CentOS) • Structured Cabling & Hardware Installation • IP CCTV, PoE, Enterprise Wi-Fi & Access Control • CIS-Style Hardening, Fail2ban & CDP / LLDP • MS 365 / Office Apps • Ticketing, SLAs & Preventive Maintenance • Remote Support (RDP / VNC / AnyDesk)

PROFESSIONAL EXPERIENCE

IT Support Engineer — **Cromptech Integrated Solutions Pvt. Ltd. · Delhi**

Dec 2024 — Present

Support end users with desktops, laptops, printers, and network devices, plus VoIP server operations, user provisioning, preventive maintenance, and LAN/WAN troubleshooting. Logs and performance are watched to hold telecom uptime at 99.9%.

PROJECTS

Dalmia Bros. — **30+ Endpoint VoIP Rollout, End to End**

July 2026 — Present | Delhi, India

I led the end-to-end VoIP deployment for Dalmia Bros., taking the project from structured cabling and hardware installation through server configuration to a live network of more than 30 IP phones. Every extension was provisioned, tested, and handed over with working call routing.

How it works: An IP desk phone is not a traditional handset. It is a small computer with a network port, a speaker, and a numeric keypad, so it needs power and a data outlet rather than a dedicated copper pair back to a phone exchange. When a call starts, the phone sends a SIP message to the call server asking to be connected. The server checks whether the number is an internal extension, a department ring group, or an outside line, then routes the call and tells both phones where to send their audio.

- Led the end-to-end installation and server configuration for the deployment.
- Connected a network of 30+ VoIP phones, with every extension tested and handed over.
- Delivered user training and post-deployment support.

TDI Bhikaji Cama Place — **Enterprise-grade IP Telephony & SIP Endpoint Provisioning**

June 2026 — Present | New Delhi, Delhi

At TDI Bhikaji Cama Place, I delivered an end-to-end IP telephony rollout that replaced legacy lines with a scalable, modern VoIP platform. I handled physical endpoint installation, VLAN segregation for voice traffic, automated extension provisioning, and a zero-downtime dial plan cutover.

How it works: DHCP options 66 and 150 are the mechanism that makes a phone configure itself. When a handset plugs into a network outlet, it asks the DHCP server for an address, and those two options tell it where to fetch its configuration file from, so nobody has to touch the handset. Quality of service works by giving voice traffic priority on the switch. Each packet carries a priority marker, and the switch forwards marked voice packets ahead of ordinary file and web traffic whenever the link gets busy.

- Implemented and configured IP phone (VoIP) systems across the organization.
- Provisioned endpoints and extensions, then verified call functionality end to end.
- Delivered post-deployment troubleshooting and user training.

Chhattisgarh Bhawan — IP CCTV, Secure Data Networking, Enterprise Wi-Fi & Biometric Access Control

January 2026 — June 2026 | Chhattisgarh, India

At Chhattisgarh Bhawan I deployed a comprehensive building-wide infrastructure covering physical security and network services together. The scope included IP CCTV surveillance, a secure segmented data network, enterprise-grade Wi-Fi coverage, and a biometric access control system, all commissioned and handed over as one coordinated build.

How it works: Power over Ethernet removes the biggest cost in camera installations. A PoE switch sends electrical power and data down the same cable, so each camera needs one outlet instead of a data point and a separate power socket, which is what makes a whole-floor camera run practical to cable. An IP camera is a small computer with a lens, so it records in digital form and sends video across the network rather than sending a signal to a tape machine. That is what makes live viewing, motion search, and remote playback possible from a phone or a browser.

- Deployed a comprehensive infrastructure covering IP CCTV, data networking, enterprise Wi-Fi, and biometric access control.
- Segmented the network so CCTV, data, and access control systems operate independently.
- Commissioned and handed over every system as one coordinated build.

LG House, Civil Lines — High-Availability Asterisk Telephony Server on Rocky Linux

May 2025 — June 2025 | Delhi, India

I engineered and deployed an enterprise IP telephony server running on Rocky Linux for LG House in Civil Lines. I configured the Asterisk IP-PBX software, provisioned more than 25 IP endpoints with hardened security rules, tuned Linux kernel networking parameters, and set up continuous log monitoring.

How it works: Asterisk is software that replaces the hardware phone exchange. A PJSIP channel manages each registered handset, RTP carries the actual audio, and a dial plan is the set of rules that decides where any dialled number should go. Because the server is a normal Linux machine, it is exposed to the same attack surface as any internet-facing host. Fail2ban watches the authentication log for repeated failed SIP logins and automatically adds the offending addresses to a block list.

- Deployed and provisioned more than 25 IP phone endpoints in an enterprise environment.
- Configured, optimized, and maintained the VoIP server on Rocky Linux.
- Monitored system logs and performance to maintain 99.9% telecom uptime.

Service Desk & Home Lab — Enterprise Network Lab & Multi-Tier SLA Service Desk

Ongoing — self-driven | New Delhi, Delhi

I designed and operate an isolated home networking and server lab that lets me test real-world enterprise architectures before they go into production. The environment simulates multi-subnet LAN/WAN routing, firewall access control lists, Active Directory domains, and a self-hosted Asterisk VoIP PBX.

How it works: A lab earns its value by letting a change be tested before it reaches a client. Anything I plan to deploy, whether a firewall rule, a subnet, or a dial plan, is built and broken on purpose here first, so the failure costs me a Saturday instead of a production incident. Two network cards on the firewall are what make the lab behave like a real site. One faces an isolated subnet standing in for the internet, the other faces the internal network, and the rules between them are the same border controls a business would deploy.

- Designed and maintained a home lab for LAN/WAN, firewall policies, and self-hosted VoIP.
- Ran ticketing and SLA workflows with remote support over RDP, AnyDesk, and VNC.
- Kept documentation, asset records, and preventive maintenance schedules up to date.

VoIP Call Quality Monitor — Packet-level call quality telemetry for a self-hosted Asterisk PBX

2025 — Ongoing | Home lab, New Delhi

Voice on an IP network fails quietly. A call does not crash when the network degrades, it simply starts sounding bad, and nobody notices until a caller complains. I built a self-hosted monitor that watches an Asterisk server in my home lab, samples the call detail records and interface counters, and scores each call so a quality drop is visible as a number rather than an opinion.

How it works: Every finished call leaves behind a call detail record, a short text line containing who called whom, how long it lasted, and the termination reason. The record is written whether the call succeeded or failed, so it is the closest thing telephony has to a log file. Packet loss, jitter, and latency are the three numbers that decide whether a call is intelligible. Packet loss is audio that never arrived and cannot be repaired in time, jitter is audio that arrived at the wrong moment, and latency is delay in the conversation. Voice codecs are built to conceal small amounts of all three, which is why a call often sounds acceptable on paper numbers that look alarming.

- Turns packet loss and jitter into a readable per-call quality score.
- Sampled on a schedule so monitoring does not become a load problem.
- Retains call history so a change can be compared against a baseline.

Service Desk Automation Bot — **Ticket triage and first-line diagnostics that cut repetitive support work**

2025 — Ongoing | Home lab, New Delhi

A large share of service desk tickets are the same handful of problems wearing different clothes. I built a triage bot that recognises those recurring patterns, runs the obvious checks automatically, and either resolves the ticket or hands it to a human with the diagnostic output already attached.

How it works: Ticket triage starts with classification: the text is matched against a set of known signatures, and the confidence of that match decides what happens next. A strong match runs its checks, a weak match is escalated, and an ambiguous one is sent to a human rather than resolved on a guess. Each signature names the checks to run rather than the answer to report. Asking for "is the print spooler running" instead of "the printer is broken" keeps the conclusion tied to evidence, which is what makes an automated resolution defensible.

- Classifies tickets and runs first-line checks before a human sees them.
- Escalates anything it cannot prove rather than guessing a resolution.
- Logs every automated action with the output that justified it.

Linux Hardening Baseline — **A repeatable Rocky and CentOS hardening script with a CIS-style audit trail**

2025 — Ongoing | Home lab, New Delhi

Hardening a server by hand is unrepeatable, and the third host drifts from the first. I wrote a baseline script that applies the same documented controls to every Rocky and CentOS machine and prints exactly what it changed, so the state of a host can be verified rather than assumed.

How it works: Idempotence is the property that makes this usable. A hardening script that has already been run changes nothing on its second pass, so it can be applied to a new host, re-applied during patching, and run as a scheduled audit without any risk of breaking a working machine. CIS benchmarks are the reference here. They are published consensus configurations rather than personal preference, which matters because a baseline someone invented for themselves is just an opinion with extra steps.

- Safe to re-run, so hardening and drift detection are the same command.
- Built on published benchmark guidance rather than personal preference.
- Reports applied, compliant, and skipped controls as separate outcomes.

Network Topology Mapper — **Automatic topology discovery and diagram generation from live switch and router data**

2025 — Ongoing | Home lab, New Delhi

Hand-drawn network diagrams go stale the moment anything moves. I built a mapper that queries the devices themselves for their neighbor tables and address plans, then generates a diagram from that live data, so the documentation is a by-product of the network rather than a separate chore.

How it works: CDP and LLDP are protocols through which network devices announce themselves to directly connected neighbors. Because a device advertises on the wire, a single query returns what is physically attached to a port, which turns discovery into a table read rather than a guess. The neighbor table gives the shape of the network but not its addressing, so interface addresses are collected alongside it. Combining the two is what lets a tool distinguish a genuine link from two devices that happen to share an address segment.

- Reads the network itself, so the diagram reflects current reality.
- Fully read-only, making it safe to run against live equipment.
- Emits a dated inventory so documentation drift becomes visible.

Backup & Restore Runbook — **Automated 3-2-1 backups with scheduled restore drills and measured RPO and RTO**

2025 — Ongoing | Home lab, New Delhi

A backup that has never been restored is a hope rather than a safeguard. I built a 3-2-1 backup setup for the lab and paired it with scheduled restore drills, so recovery time is a measured figure and the restore procedure is something I have actually performed.

How it works: The 3-2-1 rule is a shape rather than a product: three copies, on two different media types, with one kept off site. Its value is that it survives the specific failures that destroy single copies, which is a disk filling up and a site being unreachable. Backups are scheduled rather than run by hand because a manual backup is a backup that eventually gets skipped. The schedule also means the recovery point objective becomes a fact about the schedule instead of an aspiration.

- Every backup is restored on a schedule, not just written and hoped over.
- Checksums catch corruption that a file-existence check would miss.
- Recovery targets are derived from measured drills, not estimated.

AD Onboarding Automation — Scripted user lifecycle management for Active Directory onboarding and offboarding

2025 — Ongoing | Home lab, New Delhi

Creating a user in Active Directory by hand means the same sequence of clicks every time, and one missed step produces an account that is almost right. I scripted the user lifecycle so onboarding and offboarding follow the same tested path and leave a record.

How it works: Group membership decides what a user can actually reach, and it is where hand-built accounts go wrong. A new starter ends up in the wrong groups, and a leaver keeps them, so both directions are scripted from one definition of who the person is. The lifecycle is one script with modes rather than several scripts that drift apart. Onboarding and offboarding share the same source of truth for the account, so a change to that definition applies to both directions at once.

- Onboarding and offboarding run from one definition, so they cannot drift.
- Idempotent, so a step that failed midway is safe to re-run.
- Disables before deleting, keeping the identity recoverable if needed.

IT Asset Inventory Dashboard — A self-hosted web app for hardware, software, and warranty tracking

2025 — Ongoing | Home lab, New Delhi

Knowing what hardware exists, what it runs, and when its warranty expires is dull work that quietly becomes urgent. I built a self-hosted inventory dashboard that records devices, their specifications and assignments, and surfaces the ones that need attention.

How it works: A device record is worth keeping only if it can be found again, so the useful fields are the ones a support question actually arrives with: hostname, hardware model, serial, assigned user, subnet, and warranty end date. Warranty expiry is a date problem, not a record-keeping one. Calculating remaining coverage at read time means the dashboard shows what is true today rather than what was true when the row was last edited.

- Finds a device by any of the fields a support question arrives with.
- Surfaces expiring warranties and unassigned devices automatically.
- Small enough to install, read, and maintain on one host.

EDUCATION

BCA with AI Engineering — 80% — Punjabi University, Patiala

2020 — 2024

Bachelor of Computer Applications with a specialization in AI Engineering, supported by a self-built home lab for LAN/WAN, firewall, and self-hosted VoIP experiments.